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Petrologic, paleomagnetic, and structural evidence of a Paleozoic rift system in Oklahoma, New Mexico, Colorado, and Utah

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ABSTRACT

A west-northwest–trending dike swarm consisting of at least 50 mafic dikes cuts Precambrian basement rocks in the area in and south of the Black Canyon of the Gunnison, Colorado, and in Unaweap Canyon near the Colorado-Utah border. The dikes have been studied petrographically, petrochemically, and paleomagnetically, and they have been radiometrically dated. Lithologically, the dike rocks are very similar, consisting of tholeiitic diabase composed primarily of labradorite, augite, and granophyric intergrowths of quartz and alkali feldspar. A Rb-Sr isochron based on mineral separates from one dike yields an age of 495 ± 15 m.y. which is statistically identical with a previously reported Rb-Sr date. Pooling the age data for these two dikes gives an age of 497 ± 16 m.y. Whole-rock K-Ar dates on four of the dikes are scattered around the Rb-Sr date and indicate that the dikes have been relatively unaffected by reheating or hydrothermal activity since emplacement.

Paleomagnetic directional data from nine of the dikes are well grouped and consistent after both alternating-field (AF) and thermal demagnetization. Several lines of evidence indicate that the remanence reflects the dipolar geomagnetic field direction at the time the dikes were emplaced. Paleopoles corresponding to the mean directions are located at 37.0°N, 101.0°E, $\delta p = 8.6°$, $\delta m = 16.4°$ (AF demagnetization) and 37.0°N, 102.3°E, $\delta p = 4.9°$, $\delta m = 9.4°$ (thermal demagnetization). The consistency of the paleomagnetic data corroborates the idea that the dike swarm is the result of one short pulse of mantle-derived magmatic activity.

The dike swarm and other Cambro-Ordovician igneous rocks in Colorado (Powderhorn alkalic complex, Wet Mountains alkalic complexes and dikes) are aligned along a linear west-northwest trend. Cambrian diabase dikes in northeastern New Mexico and a bimodal suite (basalt–gabbro-rhyolite–granite) in southern Oklahoma (Wichita and Arbuckle Mountains, Anadarko Basin) are on line with this trend. Petrologically, all of these Cambrian and Cambro-Ordovician igneous rocks represent types that are typically associated with extensional tectonic regimes. From these data and consideration of documented Paleozoic tectonic activity along the trend of the plutons, it is concluded that throughout the Paleozoic, a west-northwest–trending tectonic zone extended from southeastern Oklahoma, through northeastern New Mexico and Colorado, and into southeastern Utah. Bimodal (basalt-rhyolite) and alkalic igneous activity was restricted to the Cambrian and Cambro-Ordovician, but tectonic activity occurred intermittently along the zone throughout the Paleozoic.

INTRODUCTION

Cambrian igneous rocks are relatively uncommon in the Midcontinent region of North America. The most notable occurrence, including spilite and rhyolite flows, gabbro, and granite, is in southern Oklahoma (see Myers and others, 1981). Cambrian volcanic rocks also occur north and west of the St. Francois Mountains, Missouri (Wagner and Kisvarsanyi, 1969; Snyder and Gerdemann, 1968), and near Provo, Utah (Stewart, 1972). Late Cambrian diabase dikes cutting Precambrian basement rocks have also been shown to exist in the northern portion of the Sangre de Cristo Range, northeastern New Mexico (Reed and others, 1983; Reed, 1984).

Cambrian and Cambro-Ordovician igneous rocks have also been identified in southwestern Colorado in the Wet Mountains, in the Powderhorn area, in the Black Canyon of the Gunnison River, and in Unaweap Canyon (Fig. 1). Alkalic rocks (including carbonatites) occur in the Wet Mountains and Powderhorn areas. West-northwest–trending diabase dikes occur in the Powderhorn area, cutting the alkalic complex, and in the Black Canyon and Unaweap Canyon areas.

Radiometric age determinations by a variety of methods indicate that the rocks in the Powderhorn alkalic complex were intruded about 570 m.y. ago (Olson and others, 1977), and those in the Wet Mountains were emplaced in 2 pulses (Olson and others, 1977; Armbrustmacher and Hedge, 1982), the alkalic complexes between 535 and 511 m.y. ago and the mafic and syenite dikes about 495 m.y. ago. Hansen and Peterman (1968) reported a Rb-Sr age of 499 m.y. for a dike in the Black Canyon area.

The purpose of this paper is to document the areal extent, petrologic nature, and age relations of the diabase dikes in southwestern Colorado and to discuss the probable tectonic setting in which they and the other Cambrian and Cambro-Ordovician igneous rocks in Colorado, New Mexico, and Oklahoma were possibly emplaced.

Extensive tables and appendices for this article may be secured free of charge by requesting Supplementary Data 85-29 from the GSA Documents Secretary.

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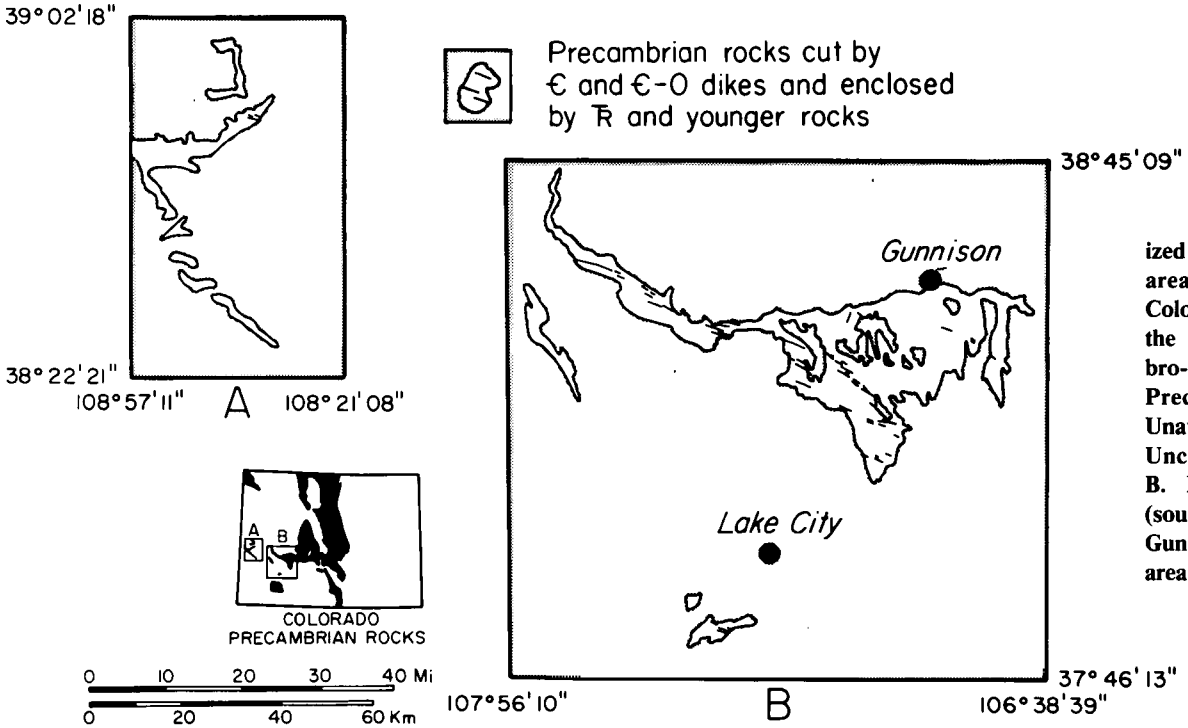


Figure 1. Generalized geologic map of two areas in southwestern Colorado (inset) showing the distribution of Cambro-Ordovician dikes in Precambrian rocks. A. Unaweap Canyon area, Uncompahgre Plateau. B. Black Canyon area (south and southwest of Gunnison) and Lake City area (south of Lake City).

DIABASE DIKES

Field Occurrence

About 50 dikes of suspected Cambro-Ordovician age, all but one trending west-northwestwardly, have been mapped in and around the Black Canyon (Hansen, 1971; Hedlund and Olson, 1973, 1974a, 1974b; Olson and Hedlund, 1973; Hedlund, 1974; Olson, 1974, 1976a, 1976b); at the southern rim of the Lake City Caldera (Lipman, 1976); and in Unaweap Canyon (Butler, 1915; Perkins, 1975), ~110 km west-northwest of the Black Canyon near the Colorado-Utah border (Fig. 1). The dikes, many of them in en echelon arrangement, are as much as 30 m wide and several kilometres long. Undoubtedly, additional dikes are present in the area, beneath the widespread Mesozoic and Cenozoic covering rocks.

We selected 14 dikes for lithologic, paleomagnetic, and radiometric studies. Collection localities and sampling and laboratory procedures are described in detail in Appendix 1 (GSA Repository).[1]

Petrography

The dikes, except for the north-northeast–trending one, are lithologically similar, and they consist of fine-grained diabase composed principally of plagioclase (~47 vol %), augite (~29 vol %), and a granophyric intergrowth of quartz and alkali feldspar (~15 vol %); iron-titanium oxides (~4 vol %), hornblende (~2 vol %), and biotite (~2 vol %) occur as minor constituents (Table 1, GSA Repository). The margins of the thicker dikes tend to be finer-grained than the central portions. Microprobe analyses of the augite, hornblende, and biotite are on file in the GSA Repository (Table 2).

Plagioclase (An_{60}) generally occurs in subhedral, tabular, unzoned grains as much as 7 mm long. Augite, which occurs in unzoned subhedral to anhedral grains as much as 3 mm across, locally has been marginally replaced by hornblende. Fine-grained granophyric intergrowths of quartz and alkali feldspar are ubiquitous. The feldspar includes both potassium feldspar and albite.

Subhedral to euhedral titanomagnetite and ilmenite are common in all thin sections. Hornblende generally occurs as a reaction product marginal to augite grains; more rarely, it occurs as subhedral primary grains. Biotite occurs both as a reaction product of augite and as discrete crystals; in some cases, it has been chloritized. Apatite is an ubiquitous accessory mineral in all samples.

In most cases, alteration of the rock has been minimal. The principal secondary phases are sericite, which most commonly replaces plagioclase in patches, and chlorite, which replaces hornblende and biotite.

Dikes at the southern margin of the Lake City Caldera were hydrothermally altered during Cenozoic volcanic activity. In these dikes, primary minerals have been largely replaced by chlorite, sericite, and hematite.

The lithology of the single north-northeast–trending dike differs from that of all of the other dikes. It lacks the granophyric quartz-alkali feldspar intergrowths and contains a large amount of chlorite (~60%), which has replaced nearly all ferromagnesian grains and most of the plagioclase. It differs markedly in chemistry (Table 3, GSA Repository), most significantly in those oxides considered to be immobile—Al_2O_3, MgO, TiO_2, and P_2O_5. For these reasons, it is considered to be Precambrian.

Petrochemistry

The chemistry of the fresh west-northwest–trending dikes is uniform throughout the area (Table 3 GSA Repository). The diabases would be classified as quartz tholeiites and K-rich tholeiitic basalts, according to the classification schemes of Yoder and Tilley (1962) and Irvine and Baragar (1971), respectively.

[1]Appendix 1 and six tables may be secured free of charge by requesting Supplementary Data 85-29 from the GSA Documents Secretary.

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Age

Until recently, the Late Cambrian age of the west-northwest–trending dike swarm was based entirely on a single Rb-Sr isochron (whole-rock and K-feldspar–rich granophyric material) from a dike in the Black Canyon area (Hansen and Peterman, 1968). In accord with the newly recommended decay constant (1.42×10^{-11} yr^{-1} for ^{87}Rb; Steiger and Jäger, 1977), this age becomes 499 ± 60 m.y. Additional dates have been reported for one of the west-northwest–trending diabase dikes and for rocks of the Powderhorn alkalic complex (40 km south of Gunnison), which are cut by the dike (Olson and others, 1977). K-Ar and Rb-Sr dates for the Powderhorn complex are concordant at 570 m.y., which is a maximum age for the transecting diabase dike. The K-Ar date on plagioclase from the dike, however, is anomalously old at 762 ± 15 m.y. (Olson and others, 1977), probably because of the presence of excess argon. Paleomagnetic data for this dike (no. 11 in Table 6, GSA Repository) are very similar to those for the other dikes, indicating that it is part of the west-northwest–trending set.

Additionally, we have done Rb-Sr analyses of mineral separates from a thick dike (no. 4) and whole-rock K-Ar analyses for samples of the freshest material available from dikes 3, 4, and 6 and from an apophysis adjacent to dike 4. Dike 4 was selected for Rb-Sr analysis because it is fresh, relatively coarse-grained, and contains a relatively high abundance of potassium feldspar.

Rb-Sr data for dike 4 (Table 4, GSA Repository), along with data for the Black Canyon diabase reported by Hansen and Peterman (1968), are shown in Figure 2. Data from dike 4, regressed according to the method of Ludwig (1982), define an age of 495 ± 15 m.y. The points scatter slightly in excess of analytical error, as indicated by a MSWD (mean square of weighted deviates) of 5.6 (if the fit is commensurate with analytical uncertainties, the value should be 1 or less). For comparison, the data of Hansen and Peterman (1968), regressed by the same technique but incorporating the larger analytical uncertainties in effect at that time, yield an age of 501 ± 33 m.y.

The ^{87}Sr/^{86}Sr intercepts for the two nearly parallel isochrons are slightly different, at $0.7053_9 \pm 0.0001_9$ and $0.7046_3 \pm 0.0006_1$ for dike 4 and the Black Canyon dike, respectively. Date and intercept errors are at the 95% confidence level.

Given that the slopes of the two isochrons are nearly identical, the two data sets have been pooled, to yield a single date, a method that involves subtraction of the ^{87}Sr/^{86}Sr intercept values of each isochron from the respective individual ^{87}Sr/^{86}Sr ratios. The uncertainties in these calculated values are increased accordingly by the uncertainties in the ^{87}Sr/^{86}Sr intercept values. The regression of these normalized data yields a date of 497 ± 16 Ma.

Of the four new K-Ar dates (Table 5, GSA Repository), two are very close to the 497-m.y. Rb-Sr age, one is too "young" by ~100 m.y., and one is too "old" by ~200 m.y. The "young" date probably is the result of argon loss, possibly related to weathering, whereas the "old" age appears to be the result of excess radiogenic argon.

On the basis of the radiometric data, particularly the close similarity of the Rb/Sr dates of two different dikes, we conclude that the best estimate of the age of the west-northwest–trending diabase dikes is 497 ± 16 m.y.—an age that is close to that of the boundary between the Cambrian and Ordovician (509 m.y.—Armstrong, 1978; and 505 m.y.—Harland and others, 1982). The K-Ar dates, although exhibiting some scatter, corroborate this age estimate and indicate that most of the dikes (with the exception of those marginal to the Lake City Caldera) have been unaffected by any significant reheating or hydrothermal event since emplacement.

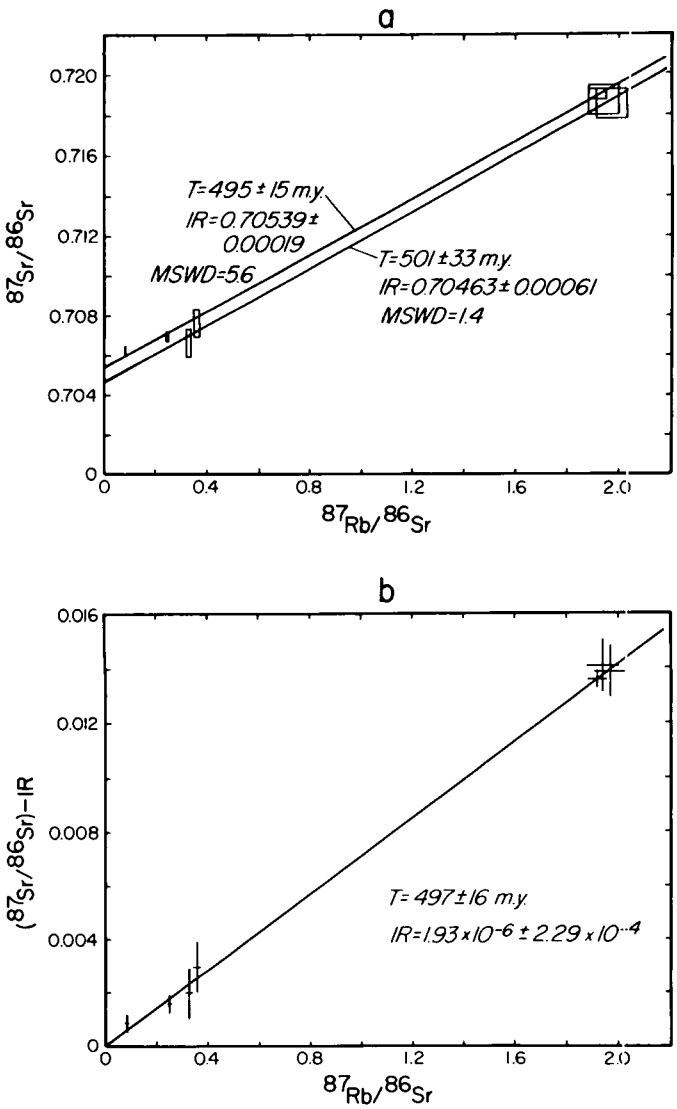


Figure 2. a. Rb-Sr isochrons determined from two different west-northwest–trending dikes: upper line, dike 4 (this study); lower line, recalculated data from Hansen and Peterman (1968). Data points for dike 4 are from plagioclase, augite, and potassium-feldspar mineral separates. Data for the isochron of dike 4 are given in Table 4 (GSA Repository). b. Rb-Sr isochron determined from the combination of data for the two diabase dikes.

Rockmagnetic and Paleomagnetic Lab Methods and Results

From each oriented hand sample from all 14 sites, 2 sets of cylindrical specimens, 2.5 cm in both length and diameter, were drilled. After measurement of natural remanent magnetization (NRM) of all specimens, 1 set was subjected to stepwise alternating-field (AF) demagnetization at peak fields of 5, 10, 15, 20, 30, and 40 mT[2], and the other set, after being given AF demagnetization at 10 mT to remove some viscous and low-field isothermal remanence components, was stepwise thermally demagnetized at 400, 500, 550, and 600 °C.

[2]For comparison, 10 mT = 100 oersteds.

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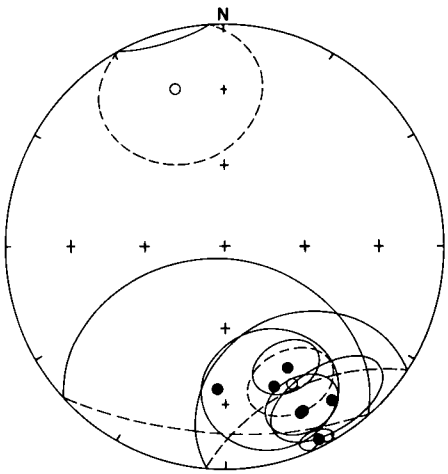


Figure 3. Mean directions of remanence after AF demagnetization for nine sites; solid circles, lower hemisphere; open circles, upper hemisphere. Cones of confidence shown by solid (lower hemisphere) or dashed (upper hemisphere) ellipses. Equal-area projection. Mean directions, cones of confidence, and other statistical data are given in Table 6 (GSA Repository).

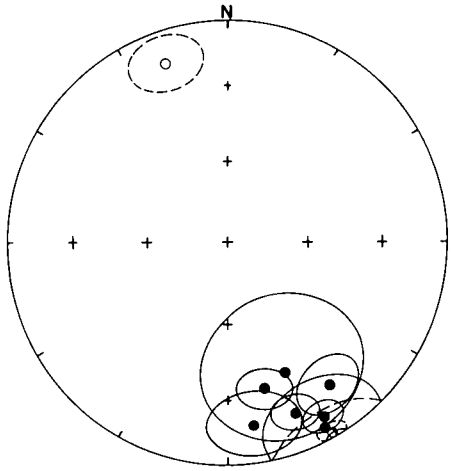


Figure 4. Mean directions of remanence after thermal demagnetization for nine sites; solid circles, lower hemisphere; open circles, upper hemisphere. Cones of confidence shown by solid (lower hemisphere) or dashed (upper hemisphere) ellipses. Equal-area projection. Mean directions, cones of confidence, and other statistical data are given in Table 6 (GSA Repository).

To determine the opaque mineralogy of the samples, polished thin sections from at least 1 specimen from each site were observed under reflected light at magnifications as great as 400×. In a few cases, polished epoxy mounts were observed first by reflected light, and again, after suitable etching in concentrated HCl, under the scanning electron microscope. In most cases, the etching brought out the fine oxidation-exsolution structures in the titanomagnetite grains. Additionally, several samples were subjected to thermomagnetic analysis.

Of the 14 dikes, 9 appear to provide "acceptable" paleomagnetic directions. Acceptance was based on the exhibition of consistent remanence-directional behavior during progressive demagnetization, on moderate to good grouping ($K > 10$) about a direction divergent from that of the modern field, and on similarity in response of a dike's mean remanence directions to both AF and thermal demagnetization techniques. Of the five that failed the acceptance criteria, two exhibited excessively strong, stable, scattered directions (apparently as a result of lightning-induced IRM); one displayed unstable scattered directions that failed to stabilize either during AF or thermal demagnetization; and two (both hydrothermally altered dikes from the southern rim of the Miocene Lake City Caldera) were completely magnetically reset during the Cenozoic. For the apparently lightning-struck samples, AF demagnetization was continued to peak fields of 100 mT, but the scatter was not reduced.

For the 9 acceptable dikes, the directions exhibited the best grouping after treatment in peak alternating fields of 20 mT or less and at temperatures of 400 or 500 °C (Table 6, GSA Repository). Means of the best-grouped directions, after both AF and thermal demagnetization, were used in calculation of the respective composite mean directions (Table 7, GSA Repository).

From observations of attitudes of Mesozoic and Cenozoic strata overlying and marginal to the Precambrian exposures, it is apparent that all dikes have been affected only by minor amounts (<4°) of tectonic rotation since the Middle Triassic. Correction for the localized rotation brings about only minor changes in the mean directions. The composite mean directions (9 dikes), after correction, for both AF and thermal demagnetization, differ from those prior to correction by about 1° (Table 7, GSA Repository). Scatter of the AF-demagnetization site means about the composite mean is little changed by structural correction, whereas that of the thermal-demagnetization site means is slightly reduced.

Inasmuch as the dike set is entirely contained within Precambrian plutonic and medium- to high-grade metamorphic basement rocks, which are unconformably overlain by Mesozoic and younger strata, it is difficult to assess directly the extent of rotation that might have affected the dikes during much of the Paleozoic. Most dikes are vertical, an orientation that one could assume is close to the original (Suppe, 1985). Those in the Black Canyon, however, dip toward the south-southwest at 75° to 80°, a fact possibly suggestive of postemplacement rotation. The mean remanence direction of the nonvertical dikes (D = 159.6°, I = 18.8°, thermal demagnetization data), however, is almost identical to that of the vertical dikes (D = 159.4°, I = 18.5°, thermal demagnetization data)—a result that supports the idea that the attitudes of the nonvertical dikes are original.

The general similarity in site mean directions of all nine dikes, both before and after structural correction (Figs. 3, 4), that were collected over a large area of southwestern Colorado suggests that the dikes have not been greatly affected by folding at any time since intrusion. There are two periods of orogenic activity that conceivably could have affected rocks of Cambro-Ordovician age in this part of Colorado—one in the middle and late Paleozoic (associated with the formation of the Ancestral Rockies) and one in the late Mesozoic–early Tertiary (Laramide Orogeny). Both of these orogenies are generally thought to be associated with vertical movement of the crust, primarily by translation along large range-margin faults (Tweto, 1975, 1980; Ross and Tweto, 1980). The well-grouped mean directions of the Cambro-Ordovician basalt dikes corroborate this conclusion and also substantiate the idea that all dikes in the swarm are of the same general age.

At thermal demagnetization levels above 550 °C, the remanence intensities sharply decreased, and the directions scattered, indicating that the principal remanence carrier resides in low-titanium magnetite. This conclusion is verified by measurement of saturation moment versus temperature: the curves are essentially reversible, blocky, and indicative of a single Curie temperature near 570 °C (Fig. 5). Under the reflecting microscope, it can be seen that nearly all original titanomagnetite grains have undergone extensive high-temperature oxidation with attendant exsolution of ilmenite lamellae along the octahedral planes. SEM observation of etched grains at 20,000× to 30,000× also reveals a finely gridded pattern parallel to the cubic (100) planes that is generally associated with relatively high-temperature subsolvus exsolution (Haggerty, 1976). The breakup of

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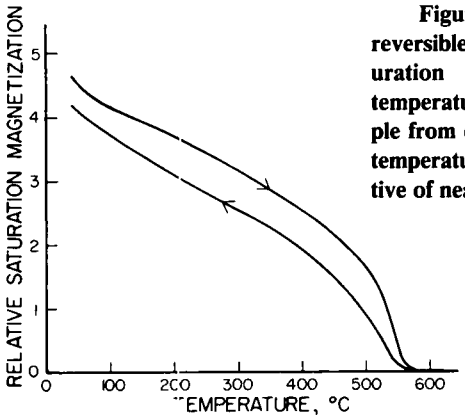


Figure 5. Typical nearly reversible thermomagnetic (saturation magnetization versus temperature) curve from a sample from dike 4. The high Curie temperature (~570 °C) is indicative of nearly pure magnetite.

the original titanomagnetite grains by ilmenite exsolution lamellae led to high magnetic stability: the mean AF demagnetization field necessary to reduce NRM intensity by one-half is ~25 mT.

Structurally corrected mean directions from AF-demagnetized samples (Fig. 3) are very similar to those from thermally demagnetized samples (Fig. 4), but it is apparent that thermal demagnetization has led to better grouping. The composite means of all 9 sites from the 2 demagnetization methods differ by <2° (Fig. 6).

Several other aspects of the paleomagnetic directional data should be stressed. First, there is a mixture of polarities: the mean direction of dike 12 is essentially antiparallel to the mean directions of the other eight dikes. This supports the idea that the composite mean direction is not that of some short-lived, nondipolar deviation in direction during Cambro-Ordovician time but probably represents the general dipolar field direction. The presence of the normal direction (northerly declination) also tends to negate the possibility that the directions are the result of remagnetization during the long-lasting (~50 m.y.), entirely reversed Kiaman magnetic interval of the late Paleozoic (Creer, 1968). Second, expected mean directions for the late Paleozoic and Mesozoic for this location are toward the southeast and *upward*; however, the mean vectors of all but one of the reversely magnetized dikes and the inverted vector of the normally magnetized dike are directed toward the southeast and *downward* (Figs. 3, 4). This strengthens the conclusion that the diabase dikes have *not* been remagnetized during the late Paleozoic and/or early Mesozoic. The fact that the remanence is primarily carried by low-titanium magnetite resulting from high-temperature oxidation-exsolution in the original titanomagnetite grains (not in hematite) also argues against remagnetization.

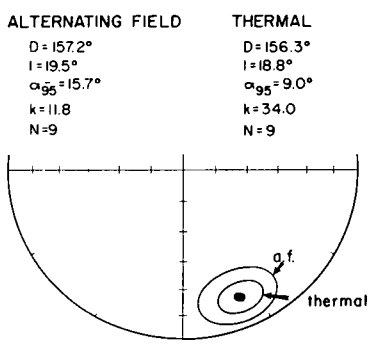


Figure 6. Compound mean directions of remanence, after both thermal and AF demagnetization, for nine dikes from the west-northwest–trending swarm and the corresponding statistical parameters. Equal-area projection. D, azimuth, clockwise from north; I, inclination, positive downward; α_{95}, half angle of the cone of confidence at the 95% level; k, precision parameter; N, number of sites.

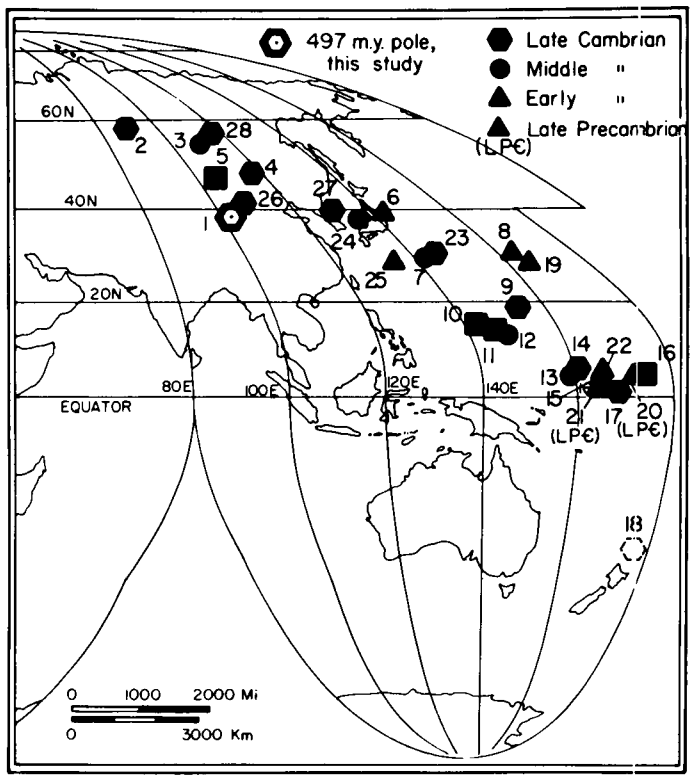


Figure 7. Plot of all available paleomagnetic pole directions for the Cambrian Period and two of late Precambrian age for reference. Dashed hexagon represents a pole at longitude 177°W. Homolosine equal-area projection. The references corresponding to the numbered poles are provided in Appendix 2 (GSA Repository). Note that the Early and Late Cambrian poles are concentrated in the low and middle latitudes, respectively.

All of the magnetic data indicate that the remanence in the west-northwest–trending diabase dikes was acquired by these rocks shortly after they formed. The mean paleomagnetic directions therefore represent valid field directions during the early Paleozoic. The results of this study moreover substantiate earlier preliminary remanence-direction results from rocks of the same dike swarm (Larson and Mutschler, 1971).

Paleomagnetic poles calculated from the structurally corrected, composite mean directions are at 37.0°N, 101.0°E (δp = 8.6°, δm = 16.4°) after AF treatment and at 37.0°N, 102.3°E (δp = 4.9°, δm = 9.4°) after thermal treatment. These other pole positions of Cambrian age of North America (Fig. 7) are confusing and somewhat contradictory (Gillett and Van Alstine, 1979; Watts and Van der Voo, 1980; Watts and others, 1980; Gillett, 1982). Cambrian pole positions are strung out, in no readily apparent, time-progressive manner, along a 45° arc that extends from an area centering on 5°N, 165°E (latest Precambrian) to an area centering on 50°N, 110°E (Ordovician and later Paleozoic).

Watts and Van der Voo (1980) and Watts and others (1980) discussed possible interpretations of the perplexing pole data. They discounted remagnetization and nondipole field variation and favored rapidly oscillating apparent pole wander. The apparent pole supposedly moved from the low-latitude position (late Precambrian) to the middle-latitude position by Middle to early Late Cambrian, then back to the low-latitude position by Late Cambrian and again back to the middle-latitude position in the Ordovician. They based the return to the low-latitude position at the

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end of the Cambrian primarily on the pole position obtained from the latest-Cambrian Point Peak Shale (Wilberns Formation), Texas, which has been dated by Rb-Sr methods on glauconite at 507 ± 12 m.y. (Morton and Long, 1978). The middle-latitude pole position we have obtained from the time-equivalent (497-m.y.) diabase dikes in southwestern Colorado (this report) presents apparently insurmountable difficulties with the looping–apparent–pole-wander model.

Gillett (1982) also recognized the difficulty in interpreting the Cambrian pole positions and concluded that some of the contradictory data must reflect remagnetization and/or tectonic rotations. It can be added that some of the problem has arisen because many of the pole positions used in compilations are based on highly scattered, inconsistent data, and the means and pole positions derived from those data are only statistical reference points without much real substance.

Following Gillett's (1982) line of reasoning, we agree that at present it is essentially impossible to construct a Cambrian pole wander path for North America that fits all of the data. For the time being, we consider that the simplest solution is to assume a relatively steady rate of apparent pole wander (corresponding primarily to a steady counterclockwise rotation of the continent) during the Cambrian, such that during the latest Precambrian the pole position is in the low latitudes and by the end of the Cambrian it is in the middle latitudes—as depicted in Figure 8.

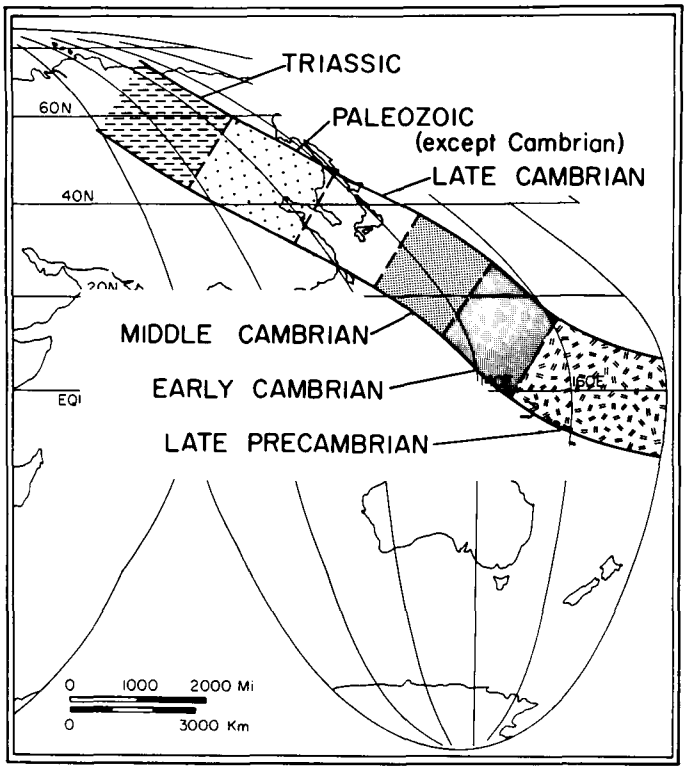


Figure 8. Simplified estimation of the pole wander path extending from the late Precambrian through the Triassic. Homolosine equal-area projection. Most of the available Cambrian pole positions (as shown in Fig. 7) are strung out along the Cambrian portion of the reconstruction.

TECTONIC SIGNIFICANCE OF THE CAMBRIAN IGNEOUS ROCKS

Of the known occurrences of Cambrian igneous rocks in Colorado, the Powderhorn complex is the oldest (570 m.y.), the Wet Mountains alkalic complexes are next oldest (535–511 m.y.), and the west-northwest–trending diabasic dike swarm in southwestern Colorado and syenite dikes in the Wet Mountains are youngest (497 and 495 m.y., respectively).

Olson and others (1977) suggested that the alkalic complexes and later dikes in the Powderhorn area and Wet Mountains represent mantle-derived plutons emplaced along a large-scale tectonic zone that possibly corresponds to the western extension of the east-trending "38th-parallel lineament" (Heyl, 1972). If, however, one considers all of the currently known occurrences of Cambrian and Cambro-Ordovician igneous rocks in Colorado (Fig. 9), it is apparent that they are aligned along a belt that parallels the strike of the west-northwest–trending Cambro-Ordovician dikes.

Along this trend, to the northwest, Cambrian basalt flows have been described from the Tintic Formation, ~40 km southeast of Provo, Utah (Stewart, 1972). These (and other mafic lavas), however, occur along the late Precambrian-Paleozoic hinge line that extended from Alaska to and beyond southern Nevada. The occurrence of the Tintic flows along the west-northwest trend of the southwest Colorado dike swarm may, therefore, be entirely fortuitous. Along this trend to the southeast (Fig. 10), Cambrian igneous rocks also occur in northeastern New Mexico (Reed and others, 1983; Reed, 1984) and in southern Oklahoma (Ham and others, 1964; Ham, 1969; Denison and others, 1966; Tilton and others, 1962; Muehlberger and others, 1966; Powell and others, 1980; Myers and others, 1981; Gilbert, 1983).

In the southern part of the Sangre de Cristo Range, northeastern New Mexico, the igneous rocks consist of regionally unmetamorphosed diabase dikes, generally trending west-northwest, which intrude Precambrian crystalline basement and are cut off by Mississippian strata. A 3-point mineral-separate Rb-Sr isochron from the thickest, freshest dike is indicative of an age near 500 m.y. (Reed, 1984).

In the Wichita and Arbuckle Mountains and in the subsurface in the Anadarko Basin, southern Oklahoma, basalt-spilite, gabbro, rhyolite, and granite, of probable Cambrian age, are abundant. The oldest rock unit is a layered gabbroic intrusion (Powell and others, 1980) that yields K-Ar dates between 730 and 509 m.y.; however, as discussed by Powell and others (1980) and Roggenthen and others (1981), these rocks may actually be Proterozoic (1,300–1,500 m.y. old), the young dates resulting from reheating during emplacement of Cambrian granitic rocks. The layered complex is cut by biotite-bearing gabbros (Powell and others, 1980) that have been dated as Middle Cambrian (Tilton and others, 1962; Burke and others, 1969). It is likely that the spilite-basalt assemblage represents the volcanic equivalent of the gabbroic intrusions, but no reliable radiometric dates have been obtained. The Late Cambrian portion of the bimodal suite (Powell and others, 1980) includes granitic plutons and thousands of metres of rhyolitic lavas and ash flows, both dated as 525± Ma. The youngest igneous rock is diabase of presumable latest Cambrian age (Gilbert, 1983). All of the plutonic and volcanic mafic rocks have tholeiitic affinities (Gilbert, 1983).

A bimodal (basalt-rhyolite) igneous assemblage such as that in southern Oklahoma is considered to be characteristic of extensional or rift-type tectonic environments (Christiansen and Lipman, 1972; Mutschler and others, 1978). This petrologic-tectonic relationship was emphasized by Hoffman and others (1974) and Gilbert (1983).

The Cambrian igneous rocks in Oklahoma are on the same trend as are those in northeastern New Mexico and Colorado (Fig. 9). Moreover, the petrogenetic association of the rocks throughout the trend is similar—mantle-derived magmas related to a rift-type (extensional) tectonic envi-

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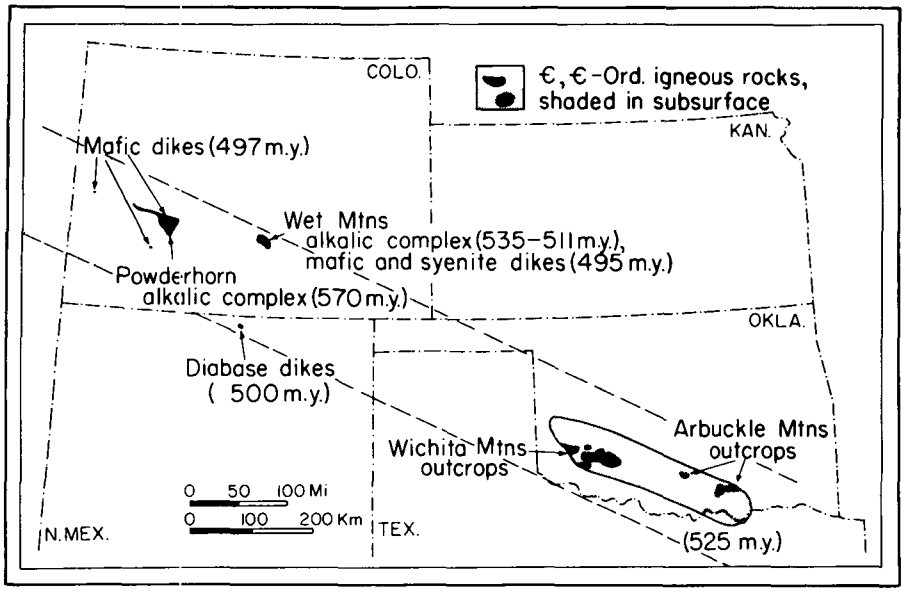


Figure 9. Distribution of Cambrian and Cambro-Ordovician igneous rocks in Colorado, New Mexico, and Oklahoma. Dashed lines indicate the general trend of their occurrence.

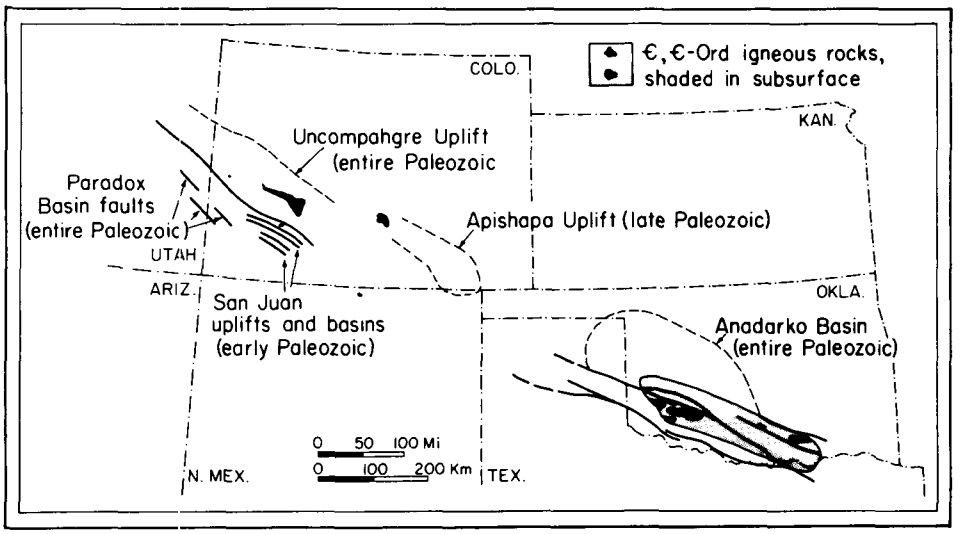


Figure 10. Distribution of Cambrian and Cambro-Ordovician igneous rocks and west-northwest-trending tectonic elements in Oklahoma, New Mexico, Colorado, and Utah. Data for Oklahoma from Ham (1967, 1969), Denison and others (1966), Tilton and others (1962), Muehlberger and others (1966); that for New Mexico from Reed and others (1983), Reed (1984); that for the Apishapa Uplift from Mallory (1972); that for the Uncompahgre Uplift, southwest Colorado, and for eastern Utah from Olson and others (1977), Mallory (1972), Baars (1965, 1972), and Baars and Campbell (1968).

ronment. It is possible, therefore, that the Cambrian igneous activity in Colorado, New Mexico, and Oklahoma was related to a single, large, west-northwest–trending tectonic zone.

As far back as 1933, Lovering and Johnson concluded that some of the unconformities in the early Paleozoic rocks along the southeastern margin of the present Front Range may have resulted from pulses emanating northwestward from tectonism in the Anadarko Basin. Gilluly (1967) postulated that the Powderhorn and Wet Mountains alkalic complexes represented igneous activity along an extension of the Anardarko-Amarillo lineament; since then, the occurrence of regional structural and igneous activity in Colorado along a west-northwest–trending zone through Oklahoma and Colorado has been repeatedly noted (Tweto, 1980; Kluth and Coney, 1981; Ross and Tweto, 1980; Ross, 1976; Baars and Stevenson, 1982).

Baars and Stevenson (1982) speculated that the Oklahoma-Colorado zone is only a part of a larger northwest- to west-northwest–trending tectonic belt that stretched from the southeastern corner of Oklahoma northwestward through and beyond western Colorado. This belt is considered to be only one of several conjugate tectonic zones crisscrossing the North American craton. They concluded that the lineaments were largely initiated during the later Precambrian (between 1,780 and 1,460 m.y. ago) and thereafter controlled the regional tectonic fabric. Baars and Stevenson based their conclusions primarily on stratigraphic-tectonic considerations from Oklahoma, Colorado, and eastern Utah.

Petrologic, stratigraphic, sedimentologic, and structural data all currently support the idea that an early Paleozoic tectonic lineament did exist, extending from Oklahoma, through northeastern New Mexico and southern Colorado, at least as far as eastern Utah. Tectonic activity was common along this trend during the Paleozoic and occurred even as recently as the Laramide (Tweto, 1975), and possibly the Quaternary (Donovan and others, 1983). The principal tectonic features, when they were active during the Paleozoic, and their spatial relations to the Cambrian and Cambro-Ordovician igneous rocks in Colorado, New Mexico, and Oklahoma are shown in Figure 10.

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Along the southeast portion of the lineament, the Anadarko Basin, which was the site of igneous activity and subsidence during the Middle Cambrian, continued to accept sediments throughout the Paleozoic. It may have started forming as early as ~1,200 m.y. ago (Gilbert, 1983). Uplift of the basin occurred during the Cambrian after emplacement of the biotite-bearing gabbros but prior to extrusion of the rhyolitic lavas during the Late Cambrian. Movement on west-northwest-trending boundary faults along the southern basin margin, which occurred throughout the life of the basin, was particularly extensive during the end stages of basin development in the late Paleozoic. There is a strong possibility that the position and trend of the Anadarko Basin was determined by earlier-formed Precambrian basement structures (Brewer and others, 1981). From COCORP deep-crustal seismic profiling, these writers found evidence supporting the existence of a deep west-northwest-trending Precambrian basin nearly coincident with the Anadarko Basin. They concluded from regional data that this basin began to develop 1,600–1,200 m.y. ago.

The Apishapa Uplift, in southeastern Colorado, which is cored by Precambrian crystalline rocks, also has a west-northwest trend. It became a source for thick, marginal, clastic red-bed sediments during late Paleozoic uplift (Mallory, 1972). Owing to the extent of this phase of erosion, and because the Apishapa Uplift is mantled by Mesozoic and later sediments, it is difficult to document early Paleozoic or Precambrian activity in and near the uplift. From well data, however, Tweto (1983) demonstrated that between 1,200 and 1,400 m.y. ago, a west-northwest–trending basin existed in southeast Colorado that accepted at least 2,000 m of sediments (Los Animas Formation). It appears to have been in existence penecontemporaneously and on trend with that in southern Oklahoma (Brewer and others, 1981).

Farther westward along the rift belt, there are several west-northwest–trending tectonic features that are certainly as old as Paleozoic (Fig. 10). Movement along high-angle faults in the Paradox Basin (southeast Utah) and uplift of the Uncompahgre structural block have been shown to have occurred repeatedly—during the Late Cambrian, the Devonian, the Mississippian, and finally during the late Paleozoic (Baars, 1965, 1966, 1972; Baars and Campbell, 1968). Linear fault-bounded uplifts and basins in southwestern Colorado have also been shown, on the basis of abrupt changes in facies and formation thickness, to have been active during the Late Cambrian and the Devonian (Baars, 1965, 1966, 1972; Baars and Campbell, 1968; Baars and See, 1968). Baars and Stevenson (1982) presented data suggesting that west-northwest–trending faults were already present in this area when the 1,460-m.y.-old Eolus Granite was intruded into 1,780-m.y.-old basement rocks.

The ages, types, and distribution of igneous rocks, as well as the trends, times of activity, and locations of structural features, indicate the presence of a large-scale west-northwest–trending rift structure extending from southeastern Oklahoma, through northeastern New Mexico and Colorado, into southeastern Utah. This structure was active during most of the Paleozoic. During the intrusion of the igneous rocks, at least, the dominant tectonic style was extensional.

Igneous activity apparently was largely restricted to a relatively short time period—the Cambrian and Cambro-Ordovician—in the development of the Oklahoma–New Mexico–Colorado–Utah tectonic zone and occurred sporadically along the zone. The nature of the magma intruded during each igneous episode was probably dependent on the depth and amount of melting that took place—greater mantle depths and smaller amounts of melting associated with the generation of the alkalic rocks and shallower mantle depths and larger amounts of melting associated with the formation of the tholeiitic rocks. The rhyolitic rocks in southern Oklahoma possibly resulted from partial melting of lower continental crust in response to heat supplied by injection of large amounts of mantle-derived mafic magma (see Hildreth, 1981).

Hoffman and others (1974) made a strong case that the Anadarko Basin represents a prime example of a failed arm of a triple junction. Gilbert (1983), however, considered the Anadarko Basin to be a rift system that shows a tectonic kinship, at least in the presence of tholeiitic igneous rocks, with either the north-trending Proterozoic North American Midcontinent rift or the northern part of the Cenozoic Rio Grande rift. When one considers the carbonatitic igneous rocks in the Colorado portion of the Oklahoma–New Mexico–Colorado–Utah tectonic zone, however, perhaps a meaningful comparison can also be made, particularly for the Cambrian, with the East African Rift Valley system, in which broadly concurrent magmatism, crustal uplift, and extensional faulting have occurred during the past 70 m.y. or so (see Best, 1982, for review).

SUMMARY

At least 50 tholeiitic dikes, as much as 30 m thick and several kilometres long, occur in a west-northwest–trending swarm cutting Precambrian rocks in and directly south of the Black Canyon of the Gunnison, in an inlier in the Lake City Caldera, and in Unaweap Canyon near the Colorado-Utah border. All are fine-grained diabase composed principally of labradorite (An_{60}), augite, and granophyric intergrowths of quartz and alkali feldspar.

A Rb-Sr isochron on mineral separates from 1 dike yields a date of 495 ± 15 Ma, which is statistically identical to the revised date (501 ± 33 Ma) of a different dike reported by Hansen and Peterman (1968). These data, when combined, give a date of 497 ± 16 Ma, which is similar to the mean whole-rock K-Ar date of the dikes. The latter indicates that the dikes have not been greatly affected by reheating or hydrothermal activity since emplacement.

Paleomagnetic directional data are consistent, after both stepwise AF and thermal demagnetization, the site mean directions differing by less than 2°. Several lines of evidence conclusively indicate that the remanence reflects the dipolar geomagnetic field at the time the dikes were emplaced. The paleopoles obtained from the dike swarm are located at 37.0°N, 101.0°E, $\delta p = 8.6°$, $\delta m = 16.4°$ (AF demagnetization) and 37.0°N, 102.3°E, $\delta p = 4.9°$, $\delta m = 9.4°$ (thermal demagnetization).

Cambrian and Cambro-Ordovician igneous rocks in Colorado, including the Powderhorn alkalic complex (570 m.y.), the Wet Mountains alkalic complexes (535–511 m.y.), diabase dikes in the area of the Black and Unaweap Canyons (497 m.y.), and mafic and syenite dikes in the Wet Mountains (495 m.y.), appear to be aligned along a linear trend that parallels that of the west-northwest–trending diabase dike swarm. Cambrian igneous rocks are also present in northeastern New Mexico, and in southern Oklahoma in the Arbuckle and Wichita Mountains and the Anadarko Basin, along a west-northwest trend that is on line with that of the Colorado intrusive rocks.

On the basis of the rock types and trend of the Cambrian and Cambro-Ordovician igneous bodies in Colorado and the Cambrian igneous rocks in New Mexico and Oklahoma, it is postulated that a large-scale west-northwest–trending rift structure extended at least from southeastern Oklahoma, through New Mexico and Colorado, into Utah. This speculation is further strengthened from a consideration of tectonic activity that occurred along the postulated trend not only during the Cambrian but sporadically throughout the Paleozoic. Activity probably began along this tectonic zone during the later part of the Precambrian, 1,600 to 1,200 m.y. ago. Igneous activity, which was sporadic, was largely restricted to the Cambrian and Cambro-Ordovician, whereas tectonic activity occurred locally and intermittently along the rift throughout the Paleozoic.

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